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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT N: 2	
TITLE: Constructors in Java	

Problem Statement

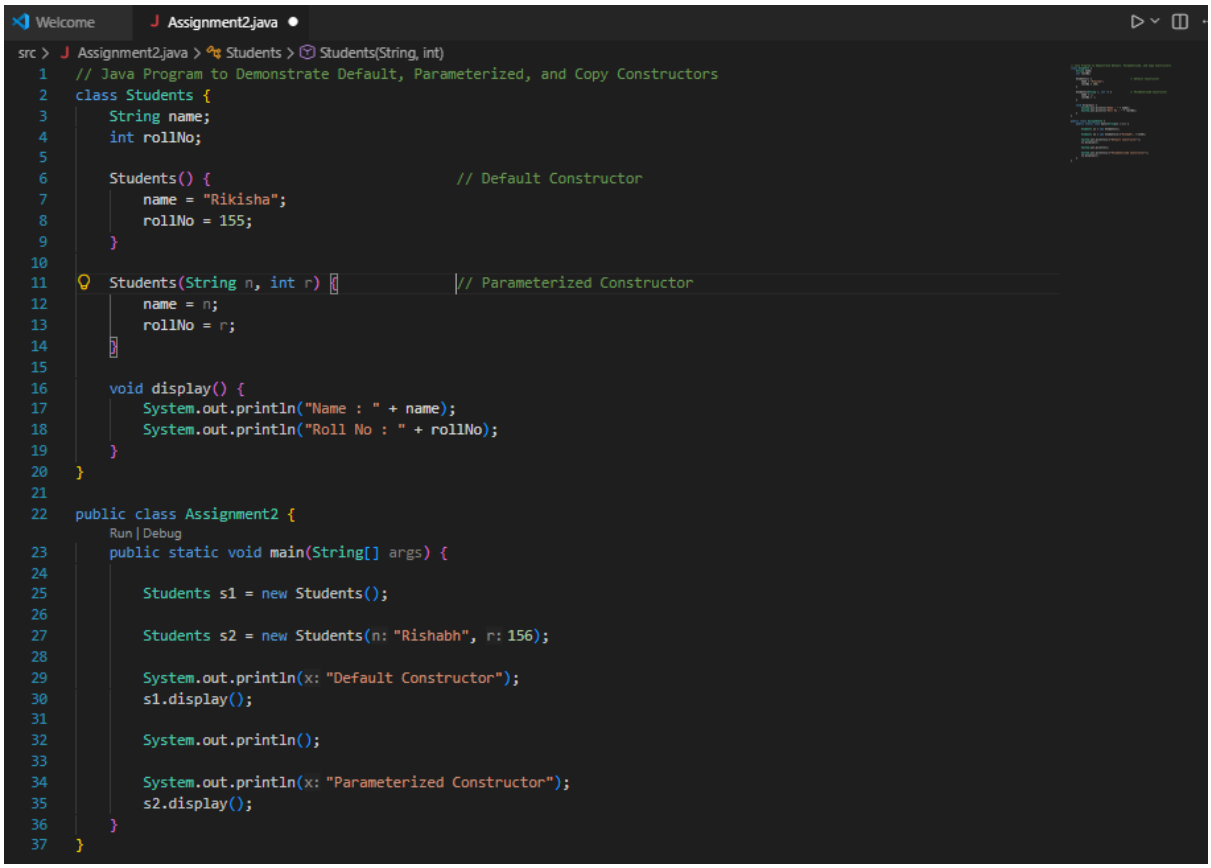
Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

1. Create a Student class using default and parameterized constructors to initialize student name and roll number.

Programme



Output

```
PS C:\Users\RIKISHA\PROGRAMMING\Java>  c:; cd 'c:\Users\RIKISHA\PROGRAMMING\Java'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:62472' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\RIKISHA\PROGRAMMING\Java\bin' 'Assignment2'
Default Constructor
Name : Rikisha
Roll No : 155

Parameterized Constructor
Name : Rishabh
Roll No : 156
PS C:\Users\RIKISHA\PROGRAMMING\Java>
```

2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

Programme

src > Assignment2.java > Mobile > Mobile(String, int)

```java
// Java Program to Demonstrate Default, Parameterized, and Copy Constructors
class Mobile {

    String brand;
    int price;

    // Default Constructor
    Mobile() {
        brand = "Samsung";
        price = 25000;
    }

    // Parameterized Constructor
    Mobile(String b, int p) {
        brand = b;
        price = p;
    }

    // Copy Constructor
    Mobile(Mobile m) {
        brand = m.brand;
        price = m.price;
    }

    void display() {
        System.out.println("Brand : " + brand);
        System.out.println("Price : " + price);
        System.out.println();
    }
}

public class Assignment2 {

    Run | Debug
    public static void main(String[] args) {

        Mobile m1 = new Mobile();

        Mobile m2 = new Mobile(b: "OnePlus", p: 35000);

        Mobile m3 = new Mobile(m2);

        System.out.println(x: "Default Constructor");
        m1.display();

        System.out.println(x: "Parameterized Constructor");
        m2.display();

        System.out.println(x: "Copy Constructor");
        m3.display();
    }
}
```

Output

```
PS C:\Users\RIKISHA\PROGRAMMING\Java>  c:; cd 'c:\Users\RIKISHA\PROGRAMMING\Java'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=t
ransport=dt_socket,server=n,suspend=y,address=localhost:52882' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\RIKISHA\PR
OGRAMMING\Java\bin' 'Assignment2'
Default Constructor
Brand : Samsung
Price : 25000

Parameterized Constructor
Brand : OnePlus
Price : 35000

Copy Constructor
Brand : OnePlus
Price : 35000

PS C:\Users\RIKISHA\PROGRAMMING\Java>
```

Conclusion

The program successfully demonstrated the use of default, parameterized, and copy constructors in Java. Constructors were used to initialize objects in different ways, making object creation more flexible and efficient.